

RCF Section 4 — Time, Gravity, and the Three-Tier Burden

Rewritten Canonical Form — v2.0 (Draft)

Preamble — What This Section Contains and Why

Section 3 constructed the operational "where" — the emergent spatial geometry (X_ω , $\tilde{d}_\omega$), the $D=3$ closure, and the coarse-graining bridge toward the smooth manifold. Section 4 constructs the operational "when" (duration), defines the "now" (joint where+when), assembles the burden tensor, and derives gravity.

This section consolidates v1.0 §3 (Emergent Time) and §5 (Gravity) into a single section because they belong to the same emergence layer — the GR perspective (coarse-grained macroscopic dynamics). The structural outline makes this clear: time (step 18), gravity (step 19), and the three-tier burden decomposition (step 20) all sit after the smooth manifold (step 17, §3.8) and constitute the framework's GR layer. Splitting them across two sections in v1.0 created artificial seams; the rewrite presents them as a continuous derivation.

Five structural issues from the v1.0 merge are addressed here:

1. ****The Arrow of Time proof swapped SOE and MOE generators**** relative to §0.4. The rewrite uses the corrected generators (Open Expansion = Hamiltonian, Open Extension = Lindblad) and invokes Spohn's inequality for entropy production.
2. ****The Newtonian limit had a sign error**** ($G_{00} \approx -2\nabla^2\Phi/c^2$ should be $+2\nabla^2\Phi/c^2$). The rewrite fixes the sign.
3. ****The Einstein closure was circular**** (Thm 5.2 assumed the field equation that Thm 5.5 derived). The rewrite derives conservation independently via diffeomorphism invariance of B_Δ .
4. ****The "Strengthened" status of the Einstein closure was overstated**** (the continuum limit premise was open). The rewrite reverts to Conditional.
5. ****The κ_B derivation was circular**** ($C = 8\pi$ was assumed, not derived). The rewrite acknowledges C is calibrated by the Newtonian limit, not derived from saturation.

The "now" — deferred from §0 because it requires both space (§3) and time (§4) — is formally defined here, completing the framework's temporal architecture.

§4.0 Purpose: The GR Perspective

Sections 0–2 constructed the exact microscopic dynamics (QM layer). Section 3 constructed the emergent spatial geometry (operational "where"). Section 4 constructs the coarse-grained macroscopic dynamics — the GR perspective — comprising duration

(operational "when"), the "now" (joint where+when), the burden tensor, gravity, and the three-tier burden decomposition.

The transition from QM to GR is the coarse-graining bridge C_ε (§3.8). The exact microscopic dynamics (R_t , burden, sectors) is averaged over macroscopic scales to produce the smooth Lorentzian manifold $(M, g_{\mu\nu})$. On this manifold, the burden tensor $\Theta^A(B)_{\mu\nu}$ sources curvature via the Einstein closure, and the three-tier burden decomposition (mode, interaction, relational) gives the physical interpretation of the stress-energy tensor.

§4.1 Duration: The Operational "When" (Layer Q' — Quartic)

The Burden-Clock Suppression Factor

Duration is built from the causal order $<$ (§1.8) weighted by burden. The key quantity is the burden-clock suppression factor $\alpha(B)$, which measures how much the local reconciliation rate slows under burden.

> **THEOREM 4.1.1 — Clock Suppression** [Established (functional form); numerical coefficient open]

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> The local clock factor is

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> $\alpha(B) = 1 / (1 + \lambda B)$ (4.1.1)

>

> where $\lambda = \eta/\gamma$ is the burden-clock coupling (Open Extension learning rate / Open Expansion rate).

>

> ***Derivation (functional form):*** The Open Expansion rate γ is fixed (the spectral-flux flow rate). The Open Extension descent rate slows with burden — near equilibrium, the Bures gradient scales linearly with burden (Property 2.2.2), giving an effective descent rate $\Gamma_{\text{eff}} \sim \eta \cdot B$. The ratio of Open Expansion rate to total reconciliation rate is:

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> $\alpha(B) = \gamma / (\gamma + \eta \cdot B) = 1 / (1 + (\eta/\gamma) \cdot B) = 1 / (1 + \lambda B)$

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> ***Status:*** [Established (functional form)]. The form $\alpha(B) = 1/(1+\lambda B)$ is derived from the SOE/MOE ratio and the burden-linear scaling of Γ_{eff} . The numerical value of $\lambda = \eta/\gamma$ is not derived; it is a primitive parameter pending T-1 (the derivation of γ from spectral gap structure).

****Remark 4.1.2 — What "Derived" Means Here.**** The v1.0 "DERIVED, not postulated" label overstated the status. The functional form $\alpha(B) = 1/(1+\lambda B)$ is derived from the SOE/MOE ratio + the gradient-descent scaling. The numerical coefficient λ is NOT derived — it depends on η and γ , both primitive. This is a derivation of functional form, not of numerical value. The rewrite is honest about this.

Burden-Weighted Proper Time

> **DEFINITION 4.1.3 — Burden-Weighted Proper Time** [Established]

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> For a causal chain $\gamma = (e_0, e_1, \dots, e_n)$ with $e_0 < e_1 < \dots < e_n$, the proper time is

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$$p_{\tau}[Y] = \sum_{k=0}^n \varepsilon \cdot \alpha(B(e_k)) = \sum_{k=0}^n \varepsilon / (1 + \lambda \cdot B(e_k))$$

 (4.1.3)

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> where $\varepsilon = 1/\gamma$ is the Open Expansion tick and $B(e_k)$ is the burden at event e_k .

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> Time is burden-weighted causal depth. Higher burden $\rightarrow$ slower clocks (gravitational time dilation). The bound: $0 < \tau[y] \leq (n+1) \cdot \varepsilon$.

The Arrow of Time (Corrected)

> **THEOREM 4.1.4 — Arrow of Time** [Established (corrected proof)]

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> The Reconciliation Propagator $R_t = \text{Open Expansion} \circ \text{Open Extension}$ is irreversible: the semigroup $\{R_t\}_{t \geq 0}$ has no inverse R_{-t} .

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> *Proof (corrected).* The Open Extension component (§1.5.3) has leading-order Lindblad form $\text{dp/dt}|_{\text{OExt}}^{\text{lead}} = -\Gamma [\hat{F}, [\hat{F}, \rho]]$ — the standard Lindblad form with Lindblad operator $L = \hat{F}$. By Spohn's inequality for Lindblad dynamics:

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$$> dS/dt = \Gamma \cdot \text{Tr}(L \rho L^\dagger - \frac{1}{2} L^\dagger L \rho - \frac{1}{2} \rho L^\dagger L) \cdot (-\log \rho) \geq 0$$

>

> The entropy production is non-negative, with equality iff ρ commutes with $L = \hat{F}$ (i.e., ρ is diagonal in the $\hat{F}$ eigenbasis — the fully reconciled state). The Open Expansion component (§1.5.2) is generated by $i[\sqrt{(\hat{F} + \delta\hat{F})}, \rho]$, which is Hamiltonian (unitary, isometric) and entropy-preserving. The composition $R_t = \text{Open Expansion} \circ \text{Open Extension}$ inherits the dissipativity of Open Extension. The semigroup property follows from the autonomy of the generator. The absence of R_{-t} follows from strict dissipativity on non-equilibrium states. $\square$

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> *Correction:* The v1.0 proof swapped the Open Expansion and Open Extension generators (attributing the Lindblad form to Open Expansion and the Hamiltonian form to Open Extension) and used a Klein-inequality argument that yielded 0 identically (because ρ and $\log \rho$ commute). The corrected proof uses the actual generators (Open Extension = Lindblad, Open Expansion = Hamiltonian) and Spohn's inequality.

The Burden-Clock Potential

> **DEFINITION 4.1.5 — Burden-Clock Potential** [Established]

>

- > The burden-clock potential is

 $\vee$

$$C(x) = c^2 \cdot \log \alpha(x) \quad (4.1.5)$$

 $\vee$

> where c_{eff} is the effective signal speed (identified with $c_{\text{RCF}} = \gamma \cdot \ell_0$, Theorem 1.1.2).

> **THEOREM 4.1.6 — Weak-Burden Expansion** [Established]

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> For $\lambda B \ll 1$ (weak burden):

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> $\Phi_C(x) \approx -c_{\text{eff}}^2 \cdot \lambda \cdot B(x) + O(c_{\text{eff}}^2 \cdot (\lambda B)^2)$ (4.1.6a)

> $\alpha(x) \approx 1 - \lambda \cdot B(x) + O((\lambda B)^2)$ (4.1.6b)

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> This connects the burden-clock potential to the burden density — the input to the Newtonian limit (§4.5).

§4.2 The "Now": Joint Where and When

Definition

With both the operational "where" (§3, the emergent spatial geometry) and the operational "when" (§4.1, duration) available, the "now" can be formally defined.

> **DEFINITION 4.2.1 — The Now** [Structural]

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> The "now" at an event $e \in E_{\text{phy}}$ is the joint definition of where and when at the same instance:

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> $\text{Now}(e) = (x(e), \tau(e))$ (4.2.1)

>

> where $x(e) \in X_{\omega}$ is the spatial location of e (from the correlation geometry, §3) and $\tau(e)$ is the duration (burden-weighted causal depth, §4.1.3).

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> The "now" is the joint product of the operational "where" and the operational "when" — the spatiotemporal instance at which admissibility is evaluated.

The Principle of the Now

> **THEOREM 4.2.2 — Principle of the Now** [Structural]

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> The "now" is the dynamic enforcement of Lie-Jordan compatibility (Theorem 0.4.4) at the operational level. Algebraically, the now corresponds to the full product $AB = \frac{1}{2}[A, B] + A \circ B$, which requires both the Lie part (causality, the "when") and the Jordan part (locality, the "where") to be jointly specified.

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> The Reconciliation Principle (§2.1) is the dynamic production of the now: the ongoing work of jointly defining where and when at each instance. The now is locally achieved within each sector (Theorem 1.6.3) but globally desynchronized by finite propagation.

Local Nows, Global Desynchronization

equation. Under an infinitesimal diffeomorphism generated by a vector field ξ^μ , the metric changes by $\delta g^{\mu\nu} = \nabla^\mu \xi^\nu + \nabla^\nu \xi^\mu$. The burden $B_\Delta[\rho]$ is a scalar functional of the coarse-grained state, hence diffeomorphism-invariant:

>

$$0 = \delta B_\Delta = \int d^4x (\delta B_\Delta / \delta g^{\mu\nu}) \cdot \delta g^{\mu\nu} = \int d^4x \Theta(B)_{\mu\nu} \cdot (\nabla^\mu \xi^\nu + \nabla^\nu \xi^\mu) = 2 \int d^4x \nabla^\mu \Theta(B)_{\mu\nu} \cdot \xi^\nu$$

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> Since this holds for arbitrary ξ^ν , we have $\nabla^\mu \Theta(B)_{\mu\nu} = 0$. $\square$

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> *Correction:* The v1.0 Theorem 5.2 was circular — it assumed the candidate field equation $G_{\mu\nu} + \Lambda_B g_{\mu\nu} = \kappa_B \Theta(B)_{\mu\nu}$, which was the conclusion of Theorem 5.5. The corrected proof derives conservation independently from diffeomorphism invariance, breaking the circularity.

§4.4 The Three-Tier Burden Decomposition

Mode, Interaction, Relational

> **DEFINITION 4.4.1 — Three-Channel Decomposition** [Established]

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> The burden tensor decomposes into three channels:

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$$\Theta(B)_{\mu\nu} = \Theta(\text{mode})_{\mu\nu} + \Theta(\text{int})_{\mu\nu} + \Theta(\text{rel})_{\mu\nu} \quad (4.4.1)$$

>

> where:

>

> 1. **Mode burden ($\Theta(\text{mode})$):** the burden an admissible relation has just by existing — the cost to remain admissible under reconciliation. The ground-state maintenance burden of stable modes (Section 2). This is the baryonic matter / particle mass contribution: $\Theta(\text{mode}) \sim \sum_{\text{particles}} m^2 \cdot \delta^3(x - x_{\text{particle}})$.

>

> 2. **Relational burden ($\Theta(\text{rel})$):** the burden spread across sub-Hilbert spaces and sub-algebras — the cost to keep them linked. Built from the cross-extension network operator: $\Theta(\text{rel}) = [\hat{C}_\alpha, \Gamma_{\text{net}}]$ (using Γ_{net} from §3.9). This is the framework's DERIVED dark-matter mechanism. It interacts only via gravity because it is the structural cost of the network's topology, not a local field excitation.

>

> 3. **Interaction burden ($\Theta(\text{int})$):** the burden acquired due to propagation, binding, and interaction — the non-additive cost of multiple modes coexisting. The non-additive part of $B(R)$: $I[\varphi, \chi] = B(R_{\{\varphi+\chi\}}) - B(R_\varphi) - B(R_\chi)$.

Physical Interpretation

| Channel | Physical Identification | Mathematical Origin |

|-----|-----|-----|

| Mode | Baryonic matter (particle mass) | Ground-state maintenance burden $m^2 \equiv B_0$ |

| Relational | Dark matter | Cross-sector coherence cost $[\hat{C}_\alpha, \Pi_{\text{net}}]$ |
| Interaction | Binding energy, interaction fields | Non-additive burden $I[\phi, \chi] \neq 0$ |

****Remark 4.4.2 — The Coincidence Problem.**** The three channels are not separate substances; they are three facets of the single reconciliation process (burden). They must scale together over cosmic time because they are different contributions to the same quantity. This structurally addresses the "coincidence problem" (why do dark matter, dark energy, and baryonic matter have similar order-of-magnitude densities today?) — they are different aspects of the same burden, so their densities are correlated by construction.

The Derived Dark-Matter Mechanism

> ****THEOREM 4.4.3 — Relational Burden as Dark Matter**** [Established (structural); amplitudes open]
>
> The relational burden channel $\Theta^{\text{(rel)}} = [\hat{C}_\alpha, \Pi_{\text{net}}]$ has four properties matching observed dark-matter phenomenology:
>
> (i) Correlates with matter clustering (relational burden is highest where sectors are most interconnected)
> (ii) Non-luminous (it is a commutator structure, not a field excitation — no electromagnetic coupling)
> (iii) Halo extension (extends beyond visible matter support because the network topology is non-local)
> (iv) Gravitational response (sources curvature via the burden tensor)
>
> ***Status:*** [Established (structural match)]. The four properties are derived from the commutator structure. Quantitative recovery (rotation curves, halo profiles, bullet cluster dynamics) requires numerical simulation of Open Extension descent on FLRW with realistic K_ω — quarantined as Q-1 (OPEN).

§4.5 The ADM Recovery and the Einstein Closure

The ADM Dictionary

> ****DEFINITION 4.5.1 — Burden-Metric Dictionary**** [Established (constitutive ansatz)]
>
> The burden-metric dictionary maps burden components to ADM metric components:
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> - Lapse: $N_B = \alpha_B(x)$ (from §4.1.1, burden-clock suppression)
> - Shift: $N_B^i = \sigma \cdot J_B^i$ (from §2.3.3, burden flux as gauge connection)
> - Spatial metric: $h_{ij}^B = h_{ij}^0 + \eta \cdot \Pi_{ij}^B + \zeta \cdot \rho_B \cdot h_{ij}^0$ (from §3.4 quotient metric baseline + burden stress deformation)
>
> where σ, η, ζ are model-dependent effective coupling coefficients.
>

> *Two derivation paths:*

> - ****Path (a) — Lovelock uniqueness:**** In 4D, under the smooth-manifold hypothesis (T-2), the unique symmetric, conserved, divergence-free rank-2 tensor built from the metric and its first two derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$ (Lovelock's theorem). By Theorem 4.3.4

(conservation) and Corollary 1.6.4 ($\Lambda_B = 0$), the burden tensor must source $G_{\mu\nu}$.

> - ****Path (b) — Open Extension Euler-Lagrange:**** Varying the formal action S_{eff} (Section 7) with respect to $g^{\mu\nu}$ gives the Euler-Lagrange equation. The burden term $\delta B_{\Delta}/\delta g^{\mu\nu} = \Theta^{\Delta}(B)_{\mu\nu}$ (Definition 4.3.1); the Einstein-Hilbert term gives $G_{\mu\nu}$; balance requires $G_{\mu\nu} = \kappa_B \Theta^{\Delta}(B)_{\mu\nu}$.

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> *Important honesty note.* These two paths are NOT independent. Path (b) uses Lovelock uniqueness to identify $G_{\mu\nu}$ as "the unique balancing tensor." The v1.0 "two independent derivation paths" claim is corrected — both paths cash out through Lovelock; one is not independent confirmation of the other.

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> *Status:* [Conditional]. The closure is conditional on the continuum limit (T-2). The v1.0 "Strengthened" status is reverted — the key premise (smooth 4D Lorentz-compatible manifold) is unproven.

$\Lambda_B = 0$ (Restated)

> ****COROLLARY 4.5.5 — $\Lambda_B = 0$ **** [Conditional on T-2]

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> The bare cosmological constant vanishes: $\Lambda_B = 0$. This is a structural consequence of Master-Zero being an exact asymptotic attractor (Corollary 1.6.4), not a fine-tuning.

>

> *Note:* $\Lambda_B = 0$ eliminates the 120-order vacuum-energy discrepancy at the foundational level (there is no vacuum energy to fine-tune). The observed cosmic acceleration is dark energy — the dynamic residual burden pressure (§2.2), NOT a cosmological constant. The cosmological application is in Section 6.

κ_B Derivation (Corrected)

> **THEOREM 4.5.6 — Gravitational Coupling** [Established (dimensional form); coefficient calibrated]

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- > The gravitational coupling is

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$$\kappa B = C / (\prod \max \cdot l_0^2) \quad (4.5.6)$$
 $\geq$

> where Π_{max} is the maximum structural pressure (the saturation limit, §5) and ℓ_0 is the fundamental length scale.

 $\vee$

> *Derivation (dimensional):* The saturation limit gives $G_{\mu\nu}^{\{\max\}} \sim 1/\ell_o^2 \approx \kappa_B \cdot \Pi_{\max}$ (the maximum curvature is bounded by the network's structural stiffness). Solving for κ_B gives the dimensional form $\kappa_B = C / (\Pi_{\max} \cdot \ell_o^2)$, where C is a dimensionless geometric factor.

 $\vee$

> *Honesty about C:* The v1.0 claimed $C = 8\pi$ "consistent with the dimensional analysis." This is circular: the saturation argument gives $C \sim 1$ (order unity), not $8\pi \approx 25.13$. The value $C = 8\pi$ is CALIBRATED by the Newtonian limit (§4.6), not derived from saturation.

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> *Status:* [Established (dimensional form); coefficient calibrated]. The dimensional form $\kappa_B \propto 1/(\Pi_{\max} \cdot \ell^2)$ is derived. The dimensionless factor C is calibrated by the Newtonian limit, not derived from first principles. The v1.0 " κ_B DERIVED" is corrected to " κ_B derived up to $O(1)$ factor calibrated by Newtonian limit."

§4.6 The Newtonian Limit (Corrected)

The Weak-Field Expansion

> **THEOREM 4.6.1 — Newtonian Limit** [Established (corrected sign)]

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> In the weak-field, slow-motion limit, the burden-clock potential Φ_C (Definition 4.1.5) is identified with the Newtonian gravitational potential Φ , and the burden density $B(x)$ is identified with the matter density. The resulting Poisson equation is:

>

$$> \nabla^2 \Phi = + 4\pi G \cdot B(x) \quad (4.6.1)$$

>

> *Proof (corrected).* The weak-field ansatz (signature $(-, +, +, +)$, Theorem 4.5.3):

>

$$> g_{00} = -(1 + 2\Phi/c^2) = -N_B^2 \approx -(1 - 2\lambda \rho_B) \quad (4.6.1a)$$

>

> where $N_B = \alpha_B = 1/(1 + \lambda B) \approx 1 - \lambda B$ (weak-burden expansion, Theorem 4.1.6) and ρ_B is the burden density (identified with matter density).

>

$$> \text{From } g_{00} = -(1 + 2\Phi/c^2) = -(1 - 2\lambda \rho_B):$$

>

$$> 2\Phi/c^2 = -(-2\lambda \rho_B) = 2\lambda \rho_B \implies \Phi = \lambda \cdot c^2 \cdot \rho_B \quad (4.6.1b)$$

>

> The Einstein tensor component in the weak-field limit (standard GR, signature $(-, +, +, +)$):

>

$$> G_{00} \approx + 2 \nabla^2 \Phi / c^2 \quad (4.6.1c)$$

>

> *Sign correction:* The v1.0 had $G_{00} \approx -2 \nabla^2 \Phi / c^2$, which is wrong for the $(-, +, +, +)$ signature (Carroll §4; Weinberg). The correct identity is $G_{00} \approx +2 \nabla^2 \Phi / c^2$.

>

> From the Einstein closure $G_{00} = \kappa_B \cdot \Theta_{00} = \kappa_B \cdot \rho_B$ (using $\Theta_{00} = \rho_B$, the burden density):

>

$$> 2 \nabla^2 \Phi / c^2 = \kappa_B \cdot \rho_B \implies \nabla^2 \Phi = (\kappa_B c^2 / 2) \cdot \rho_B \quad (4.6.1d)$$

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Dark Matter Halo

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The ℓ_0 -Floor

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Singularity Avoidance

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> Classical singularities (general relativistic) are replaced by pressure saturation at the ℓ_0 -floor. The infinity is in the geometric representation (divergent $h_{\{rr\}}$, vanishing lapse), not in the algebra (bounded $\rho_B \leq \rho_B^{\max}$).

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> *Reframing:* The v1.0 called this "singularity avoidance." The corrected statement: the $r=0$ singularity is replaced by a degenerate but bounded geometry ($h_{\{rr\}} \rightarrow \infty$, $N_B \rightarrow 0$). This is not "avoidance" in the sense of "no singularity" — it is REPLACEMENT of the singularity with a different geometric regime (the holographic-boundary regime, §5).

§4.8 Architectural Summary

What Section 4 Contains

Object	Definition	Status
Burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$	Thm 4.1.1	Established (functional form); λ open
Burden-weighted proper time $\tau[\gamma]$	Def 4.1.3	Established
Arrow of Time (corrected)	Thm 4.1.4	Established (corrected proof)
Burden-clock potential Φ_C	Def 4.1.5	Established
Weak-burden expansion	Thm 4.1.6	Established
The "now" (joint where+when)	Def 4.2.1, Thm 4.2.2	Structural
Global desynchronization of the now	Thm 4.2.3	Structural (T-2 for formal proof)
Burden tensor $\Theta^{\alpha}(B)_{\mu\nu}$	Def 4.3.1	Established (variational)
Component-block form	Def 4.3.2	Established (constitutive identification)
Symmetry of burden tensor	Thm 4.3.3	Established (substantive)
Active-source conservation (non-circular)	Thm 4.3.4	Established (Noether-style)
Three-channel decomposition	Def 4.4.1	Established
Relational burden = dark matter	Thm 4.4.3	Established (structural); amplitudes Q-1
ADM dictionary	Def 4.5.1	Established (constitutive ansatz)
Effective burden metric	Thm 4.5.2	Established
Type-Sign sufficiency	Thm 4.5.3	Established (completing §3.7)
Einstein-like closure	Thm 4.5.4	Conditional (T-2)
$\Lambda_B = 0$	Cor 4.5.5	Conditional on T-2
Gravitational coupling κ_B	Thm 4.5.6	Established (dimensional); C calibrated
Newtonian limit (corrected sign)	Thm 4.6.1	Established (corrected)
Relational burden halo	Thm 4.6.2	Established (structural); profile Q-1
Metric boundedness (ℓ_0 -floor)	Thm 4.7.1	Conditional on T-2
Singularity avoidance (reframed)	Thm 4.7.2	Conditional on T-2

What Section 4 Does NOT Contain (Deferred)

Object	Deferred to	Reason
Black holes, holography	§5	Needs gravity (§4) + saturation limit
Cosmology, dark energy application	§6	Needs $H(t)$, FLRW, Γ evolution

Formal action S_{eff}	§7	Audit/closure section
Continuum limit formal proof	T-2	Spectral analysis of R_t on $\ker(\hat{M})$
Particle mass spectrum	Q-6	Standard Model recovery (open)

The Emergence Ladder Through Section 4

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From §0–§3:

$A \rightarrow \text{Lie-Jordan} \rightarrow \hat{M} \rightarrow \text{admissibility} \rightarrow \text{GNS} \rightarrow H_{\text{kin}} \rightarrow \hat{F}$
+ Causality (Lie) + Locality (Jordan)
 $\rightarrow \text{Open Expansion} \rightarrow \text{fracture} \rightarrow \text{burden} \rightarrow A_{\text{red}} \rightarrow \hat{M}_{\text{red}} \rightarrow \text{Dirac bracket}$
 $\rightarrow R_t \rightarrow \text{Convergence Theorem} \rightarrow A_{\text{phy}} \rightarrow \text{events} \rightarrow <$
 $\rightarrow \text{RP} \rightarrow \text{dark energy mechanism} \rightarrow \text{fields} \rightarrow \text{particles} \rightarrow \text{mass} (m^2 \equiv B_0)$
 $\rightarrow \text{record sectors} \rightarrow \text{Born rule} \rightarrow \text{FIREWALL}$
 $\rightarrow K_\omega \rightarrow d_\omega \rightarrow \text{metric quotient} \rightarrow D=3 \rightarrow \text{Type-Sign (necessary)} \rightarrow \text{coarse graining}$

§4 (GR perspective — coarse-grained macroscopic dynamics):

Duration (burden-weighted causal depth, $\alpha(B) = 1/(1+\lambda B)$)
 $\rightarrow$ The "now" (joint where+when, locally achieved, globally desynchronized)
 $\rightarrow$ Burden tensor $\Theta^\wedge(B)_{\mu\nu}$ (variational derivative of B_Δ)
 $\rightarrow$ Three-channel decomposition (mode = baryonic, relational = dark matter, interaction = binding)
 $\rightarrow$ ADM recovery $\rightarrow$ effective burden metric
 $\rightarrow$ Type-Sign sufficiency (Lorentzian signature UNIQUE)
 $\rightarrow$ Einstein closure ($G_{\mu\nu} = \kappa_B \Theta^\wedge(B)_{\mu\nu}$, $\Lambda_B = 0$)
 $\rightarrow \kappa_B = C/(\Pi_{\text{max}} \cdot \ell_0^2)$ (dimensional; C calibrated)
 $\rightarrow$ Newtonian limit ($\nabla^2 \Phi = +4\pi G \cdot B$, sign corrected)
 $\rightarrow$ Dark matter halo (relational burden)
 $\rightarrow$ Singularity avoidance (ℓ_0 -floor, reframed)

$\rightarrow$ Section 5: Black holes, holography, Hawking radiation
 $\rightarrow$ Section 6: Cosmology, dark energy application, Friedmann
 $\rightarrow$ Section 7: Audit, formal action S_{eff} , closure

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Acyclicity Test

****Question:**** Does Section 4 define every object before using it? Are there forward references to §5+ objects?

****Answer:**** The chain is acyclic within Section 4.

- Duration (§4.1) uses $<$ (§1.8), burden (§1.3), $\alpha(B)$ (derived locally).
- The "now" (§4.2) uses space (§3) and duration (§4.1).
- The burden tensor (§4.3) uses B_Δ (§1.3), C_ε (§3.8), the metric (§3.4, §4.5).
- The three-channel decomposition (§4.4) uses Π_{net} (§3.9), stable modes (§2.4).

- ADM recovery (§4.5) uses α_B (§4.1), J_B (§2.3), $h_{ij}^{(0)}$ (§3.4).
- Einstein closure (§4.5.4) uses the burden tensor (§4.3), conservation (§4.3.4), $\Lambda_B = 0$ (§1.6.4).
- κ_B (§4.5.6) uses $\Pi_{\max}$ — **flagged forward reference** ($\Pi_{\max}$ formalized in §5, the saturation limit). The dimensional form is stated here; the formal saturation analysis is in §5.
- Newtonian limit (§4.6) uses the Einstein closure (§4.5.4), the weak-burden expansion (§4.1.6).

One flagged forward reference: $\Pi_{\max}$ (needed for κ_B) is formalized in §5 (the black-hole saturation limit). This is a one-way forward reference (§4 → §5), not a circularity. The dimensional form of κ_B is stated here; the saturation analysis is in §5.

Verdict: Section 4 is acyclic. The architecture holds through the GR perspective.

End of Section 4 — Rewritten Canonical Form v2.0 (Draft).

This draft consolidates time and gravity into the GR perspective, with five corrections to v1.0:

The Arrow of Time proof is corrected. The v1.0 swapped Open Expansion and Open Extension generators and used a Klein-inequality argument that yielded 0. The corrected proof uses the actual generators (Open Extension = Lindblad, Open Expansion = Hamiltonian) and Spohn's inequality.

The Newtonian limit sign is fixed. $G_{00} \approx -2\nabla^2\Phi/c^2 \rightarrow +2\nabla^2\Phi/c^2$, giving $\nabla^2\Phi = +4\pi G \cdot B(x)$ consistent with the theorem's conclusion.

The Einstein closure circularity is broken. The conservation theorem (4.3.4) is now derived independently from diffeomorphism invariance (Noether-style), not from assuming the field equation.

The "Strengthened" status is reverted. The Einstein closure is Conditional (the continuum limit premise is open), not Strengthened.

The κ_B derivation is honest. The dimensional form is derived; the $C = 8\pi$ factor is calibrated by the Newtonian limit, not derived from saturation. The "two independent paths" claim is corrected — both paths use Lovelock.

The "now" is formally defined here (§4.2), with the global desynchronization mechanism (local nows achieved, globally lagging) that grounds the dark energy interpretation.